Circuit Theory Foundations — Reference Encyclopedia

Sections 1–13: Charge, Current, Voltage, Resistance, Circuits,
and Systematic Analysis — Complete Edition

Definitions, physical mechanisms, derivations, and fully worked examples, cross-referenced to
ECE 105 Lectures 5–7 (Circuit Analysis 1–3)

ECE 105 — Introduction to Electrical Engineering

August 10, 2026

Contents

Chapter 1

Introduction and How to Use This Document

This is the complete, self-contained reference for the first unit of ECE 105: the foundations of circuit theory. It fuses the course-notes sections 1–13 with the material actually presented in **Lecture 5 (Circuit Analysis 1)**, **Lecture 6 (Circuit Analysis 2)**, and **Lecture 7 (Circuit Analysis 3)**. Where the notes and the lectures overlap, the fuller treatment is kept; where the lectures add material not in the notes (nodal analysis, capacitors, inductors), it is folded in. Each section states its governing equations explicitly, explains the physical mechanism behind them, and works numerical examples in full, so you can jump straight to whatever you are stuck on.

Structure. **Part I** — the physical foundation: charge, current, voltage, and power, including *why* charge moves and what voltage really is. **Part II** — resistance, resistivity, conductivity, and Ohm's law, including the geometry and material dependence of resistance. **Part III** — circuit building blocks and reduction: series and parallel, the dividers, complex-network reduction, the loaded divider and the 10-percent rule, real sources, combining batteries, measurement, and faults. **Part IV** — systematic analysis for networks that will not reduce: Kirchhoff's laws (with the determinant/Cramer method), nodal analysis, superposition, and the Thévenin/Norton equivalents. **Part V** — the energy-storage elements introduced at the end of the unit: capacitors and inductors. Appendix A is a consolidated formula sheet.

Conventions. Current is *conventional current* — the direction positive charge would move — which is opposite to the real drift of electrons in a metal. Every formula here uses conventional current; the electrons' true direction never enters the algebra. Unless noted, resistances are in ohms, voltages in volts, currents in amperes. Voltages and energies are always *differences between two points*, even when written with a single symbol.

Part I

The Physical Foundation

Chapter 2

What Circuit Theory Describes

Electrical engineering is built on five circuit quantities, introduced on the opening slide of Lecture 5:

Quantity	Symbol	Unit
Voltage	V	volt (V)
Current	I	ampere (A)
Resistance	R	ohm (Ω)
Capacitance	C	farad (F)
Inductance	L	henry (H)

Circuit theory is the set of rules that predicts the voltage across, and current through, every element of a network built from these quantities. Everything in this unit is one of three things: a *definition* of a quantity (current, voltage, resistance), a *law* relating them (Ohm, Kirchhoff), or a *technique* that applies those laws to a whole network (series/parallel reduction, nodal analysis, Thévenin/Norton).

Key idea A circuit problem is "solved" when every branch current and every node voltage is known. Every method in Part IV is a different route to that same goal — reduction when the circuit is simple, Kirchhoff or nodal analysis when it is not, Thévenin/Norton when only one pair of terminals matters.

The mathematics required is almost entirely algebra. Calculus appears in only two places — the instantaneous definitions $I = dQ/dt$ and $I = C\,dV/dt$ — and even there you rarely compute a derivative by hand.

Chapter 3

Electric Charge and Current

3.1 Charge

Electric charge Q (unit: coulomb, C) is the fundamental property carried by the constituents of atoms. In a metal such as copper, each atom donates roughly one **free electron** to a shared "sea" that drifts through a lattice of fixed positive ions. The elementary charge is

$$e = 1.602 \times 10^{-19}\,\mathrm{C}, \tag{3.1}$$

with the electron carrying $-e$ and the proton $+e$. A conductor is electrically neutral overall — equal numbers of electrons and protons — even while a current flows through it.

3.2 Current as the rate of charge flow

Electric current is the total charge passing through a cross-section of the conductor per unit time. If ΔQ passes in time Δt, the average current is $I_{\mathrm{ave}} = \Delta Q/\Delta t$; taking the limit gives the instantaneous current:

$$I = \lim_{\Delta t \to 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}, \qquad 1\,\mathrm{A} = 1\,\mathrm{C/s}. \tag{3.2}$$

The ampere ("amp") is a large unit; currents are often quoted in milliamps ($1\,\mathrm{mA} = 10^{-3}\,\mathrm{A}$), microamps ($10^{-6}\,\mathrm{A}$), or nanoamps ($10^{-9}\,\mathrm{A}$).

Worked Example 3.2.1 (electrons behind a given current)

How many electrons pass a point in 3 s when the current is 2 A? The charge is $\Delta Q = I\,\Delta t = (2)(3) = 6\,\mathrm{C}$, so the number of electrons is

$$\frac{6\,\mathrm{C}}{1.602 \times 10^{-19}\,\mathrm{C}} = 3.74 \times 10^{19}.$$

Worked Example 3.2.2 (current from a time-varying charge)

If $Q(t) = (0.001\,\mathrm{C})\sin(1000\,t/\mathrm{s})$, then by Eq. (3.2)

$$I = \frac{dQ}{dt} = (0.001)(1000)\cos(1000\,t) = (1\,\mathrm{A})\cos(1000\,t).$$

At $t = 1\,\mathrm{s}$ this is $0.174\,\mathrm{A}$; at $t = 3\,\mathrm{s}$ it is $-0.5\,\mathrm{A}$, the minus sign meaning the current momentarily flows the other way — the hallmark of an alternating (AC) quantity.

3.3 Conventional current versus electron flow

When Benjamin Franklin named the moving charges, he assumed they were positive. J. J. Thomson later showed the movers in a metal are actually negative electrons, drifting *opposite* to Franklin's assumed direction — but by then all the formulas were written in Franklin's convention.

Key idea **Conventional current** I points the way positive charge would move. In a metal the electrons actually drift the opposite way. Negative charge moving left is electrically identical to positive charge moving right, so every formula (Ohm's law, the power law, KCL) works perfectly while "pretending" current is positive. When you read "electron flow," picture conventional current pointing the other way.

3.4 Drift velocity, and currents in perspective

The electrical "signal" propagates near the speed of light — flip a switch and the whole loop responds almost at once — but the electrons themselves crawl. The **drift velocity** (their net progress toward the positive terminal) is astonishingly slow: about $0.002\,\mathrm{mm/s}$ for a $0.1\,\mathrm{A}$ current in a 12-gauge wire. The fast pulse is the repulsive "shove" passing down the electron sea; the slow drift is the sea's bulk motion.

For a sense of scale: a typical LED draws about $20\,\mathrm{mA}$; a smartphone on the web, $\sim 200\,\mathrm{mA}$; a $100\,\mathrm{W}$ bulb, about $1\,\mathrm{A}$; a toaster, 7–10 A; a car starter, $\sim 200\,\mathrm{A}$; a lightning strike, $\sim 1000\,\mathrm{A}$. Currents from roughly $100\,\mathrm{mA}$ to $1\,\mathrm{A}$ through the body can be fatal — respect even "small" currents.

Chapter 4

Voltage, EMF, and Power

4.1 What voltage is

To make current flow between two points, a **voltage** must exist between them. A voltage sets up an **electromotive force** (EMF) that pushes on every free electron in the conductor. “Voltage,” “potential difference,” and “potential” all name the same thing; we avoid “potential” alone because it is easily confused with potential *energy*, which is different.

4.2 The mechanism: electrical pressure

Inside a battery, chemical reactions pile up electrons at the negative terminal. Electrons repel one another, so this pile is a region of high “electrical pressure.” Connect a load and the crowded electrons shove outward into the circuit; the shove races around the loop near light speed even though any individual electron barely moves. Far “downstream” — deep inside a lamp filament, say — the shove is weaker because collisions along the way have dissipated energy. Regions of weak shove are regions of low voltage: the electrons there have little ability to do further work. The battery and the circuit need each other — without a path between the terminals the internal chemistry, which transfers electrons through the external link, cannot proceed.

Key idea Voltage is the difference in potential energy per unit charge between two points:

$$V = \frac{U}{q}, \qquad 1\,\mathrm{V} = 1\,\mathrm{J/C}.$$

Two points 1 V apart can do 1 J of work for every 1 C that moves between them. A 1.5 V battery does 1.5 J of work per coulomb it pumps around the circuit.

Worked Example 4.2.1 (energy carried by the charge)

Across a 1.5 V cell, one electron ($q = 1.602 \times 10^{-19}\,\mathrm{C}$) carries a potential-energy difference $\Delta U = \Delta V\, q = (1.5)(1.602 \times 10^{-19}) = 2.4 \times 10^{-19}\,\mathrm{J}$. If the flashlight draws 0.1 A — about 6.24×10^{17} electrons per second — the energy delivered per second is about $0.15\,\mathrm{J/s} = 0.15\,\mathrm{W}$, which is exactly the power we compute below.

4.3 Ground, point voltage, and the $0\,\mathrm{V}$ reference

Because voltage is always *between two points*, giving a single node "a voltage" requires a reference. We pick one point — usually the negative terminal of the supply — call it $0\,\mathrm{V}$, mark it with a **ground** symbol, and quote every other node's **point voltage** relative to it. Within a single ideal conductor there is no voltage difference at all: a voltmeter placed across two points of the same wire reads $0\,\mathrm{V}$ (in reality a few μV, from the wire's tiny internal resistance).

4.4 Power and the generalized power law

Power is energy per unit time. Substituting $U = Vq$ (with V constant) into $P = dU/dt$ and using $I = dq/dt$:

$$P = \frac{dU}{dt} = V\frac{dq}{dt} = VI, \qquad 1\,\mathrm{W} = 1\,\mathrm{J/s} = 1\,\mathrm{V}\cdot\mathrm{A}. \tag{4.1}$$

This is the **generalized power law**. It is "generalized" because it holds for *any* two-terminal element regardless of material or mechanism, and it gives a second definition of the volt: $1\,\mathrm{V} = 1\,\mathrm{W/A}$. It tells you *how much* power an element handles, but not *how* the power is used — that requires knowing the element (a resistor turns it to heat; a motor, to motion).

Worked Example 4.4.1 (the two everyday uses of $P = VI$)

(a) A $1.5\,\mathrm{V}$ flashlight drawing $0.1\,\mathrm{A}$ consumes $P = (1.5)(0.1) = 0.15\,\mathrm{W}$. (b) A $12\,\mathrm{V}$ device rated at $100\,\mathrm{W}$ draws $I = P/V = 100/12 = 8.3\,\mathrm{A}$.

4.5 Combining batteries and the split supply

Two equal cells in *series* double the electrical pressure: their voltages add (two $1.5\,\mathrm{V}$ cells give $3.0\,\mathrm{V}$) because you have put two charge pumps in line. Choosing where to place the $0\,\mathrm{V}$ ground reference sets the point voltages: grounding the bottom terminal of a $3.0\,\mathrm{V}$ stack gives point voltages of 0, 1.5, and $3.0\,\mathrm{V}$; moving the ground to the *middle* yields $+1.5\,\mathrm{V}$ and $-1.5\,\mathrm{V}$ rails — a **split supply** whose ground is a *common return*. Circuits with signals that swing both above and below zero (audio, for instance) need exactly this.

4.6 Sources of EMF other than batteries

Chemical action is one of several EMF mechanisms. *Magnetic induction* (generators) and *photovoltaic action* (solar cells) are the only others powerful enough to run most circuits. The *thermoelectric* and *piezoelectric* effects produce millivolt-level signals useful for sensing but not for power. *Static electricity* can reach thousands of volts but delivers only a brief, often destructive, pulse — a nuisance, not a power source.

4.7 Water analogies

It often helps to picture a voltage source as a pump, wires as pipes, (Franklin's positive) charge as water, and current as flow. A load is a constriction that limits the flow. A gravity version — a raised tank whose height sets the "pressure" — captures the key intuition that *more voltage drives more current*, though like all analogies it eventually breaks down.

Part II

Resistance and Ohm's Law

Chapter 5

Resistance, Resistivity, Conductivity

5.1 Ohm's definition of resistance

As electrons drift, they collide constantly with lattice ions, impurities, and one another; these collisions impede the flow, and we call the net impediment **resistance**. In 1826 Georg Simon Ohm measured, for many materials, a linear relation between the voltage applied and the current that resulted, and *defined* resistance as their ratio:

$$R \equiv \frac{V}{I}, \qquad 1\,\Omega = 1\,\mathrm{V/A}. \tag{5.1}$$

5.2 Ohmic and nonohmic materials

An **ohmic** material has (nearly) constant R over a range of voltages, so its I–V graph is a straight line through the origin and $V = IR$ holds. A **nonohmic** element — a diode, a lamp filament — has an R that changes with operating point; its I–V curve bends, and $V = IR$ (with constant R) does not describe it. A diode, for example, passes current easily one way and blocks it the other.

Key idea Ohm's law is not a fundamental law of nature; it is the empirical observation that *some* materials have constant resistance. The *definition* $R = V/I$ always holds, point by point, for any element. "$V = IR$ with R constant" holds only for ohmic elements. Note the logic: you do not define voltage from current and resistance — you define *resistance* from voltage and current. (You may of course rearrange $V = IR$ to predict a voltage from a known resistance and measured current, which circuit analysis does constantly.)

5.3 How geometry sets resistance

For a uniform conductor, resistance grows with length and shrinks with cross-sectional area. Doubling the length doubles R (twice as many collisions to fight through); doubling the area halves R (the same current spreads over more room, so fewer collisions). The concentration of current is the **current density**

$$J = \frac{I}{A} \quad (\mathrm{A/m^2}), \tag{5.2}$$

which is larger, and the drift velocity higher, in a thin wire than a thick one carrying the same current.

5.4 Resistivity and conductivity

Geometry aside, the *material* matters. **Resistivity** ρ is the shape-independent property:

$$R = \rho\,\frac{L}{A}, \qquad \rho \equiv R\,\frac{A}{L}\ (\Omega{\cdot}\mathrm{m}), \qquad \sigma \equiv \frac{1}{\rho}\ (\mathrm{S}), \tag{5.3}$$

where σ is the **conductivity** (siemens). Resistivity and conductivity carry the same information — "glass half empty" versus "half full." In these terms Ohm's law reads $V = \rho\,(L/A)\,I = IL/(\sigma A)$. Metals such as copper and silver have conductivity a factor of $\sim 10^{21}$ above a good insulator like Teflon. Silver is the best common metal but too costly; aluminum conducts well but oxidizes into a fire hazard in house wiring, so copper dominates.

Worked Example 5.4.1 (resistance of a length of copper wire)
Copper: $\rho \approx 1.7 \times 10^{-8}\,\Omega{\cdot}\mathrm{m}$. For $L = 10\,\mathrm{m}$ and $A = 1\,\mathrm{mm}^2 = 1 \times 10^{-6}\,\mathrm{m}^2$,

$$R = \rho\frac{L}{A} = 1.7 \times 10^{-8} \cdot \frac{10}{1 \times 10^{-6}} = 0.17\,\Omega.$$

This is why short runs of wire are treated as ideal $0\,\Omega$ links.

5.5 Temperature dependence

Over a moderate range, most metals follow

$$\rho = \rho_0\big[\,1 + \alpha(T - T_0)\,\big], \tag{5.4}$$

where α is the **temperature coefficient of resistivity** ($^\circ\mathrm{C}^{-1}$). Metals grow *more* resistive when hot because lattice vibrations scatter the drifting electrons more violently.

5.6 Are water, air, and vacuum conductors?

Pure water is a good insulator ($\rho \approx 2.5 \times 10^5\,\Omega{\cdot}\mathrm{m}$); it conducts only via the sparse H_3O^+/OH^- ions of autoionization. *Salt water* conducts well ($\rho \sim 0.2\,\Omega{\cdot}\mathrm{m}$) because dissolved Na^+ and Cl^- act as carriers. *Air* is normally an insulator, but a strong enough field (breakdown $\sim 3\,\mathrm{MV/m}$ between plane electrodes) rips electrons free and ionizes it, producing three classic discharges:

- **Corona discharge:** ionization confined to a small region around a sharp electrode where the field exceeds breakdown; a faint, sustained leakage current.
- **Spark discharge:** a fast, full breakdown all the way between two rounded conductors — the "zap" after scuffing across a carpet. People rarely feel discharges below $\sim 1000\,\mathrm{V}$, notice discomfort around $2000\,\mathrm{V}$, and object above $3000\,\mathrm{V}$.
- **Brush discharge:** intermediate between the two — the crackle of pulling off a sweater. Discharges from insulators are almost always brush discharges.

A *vacuum* conducts only if charges are injected into it (as in a vacuum tube), since there are no carriers to begin with.

Chapter 6

Ohm's Law, the Power Law, and Resistors

6.1 The two working equations

Combining Ohm's law $V = IR$ with the generalized power law $P = VI$ gives the **resistor power law** in three interchangeable forms:

$$P = VI = I^2R = \frac{V^2}{R}. \tag{6.1}$$

Use whichever two quantities you already know.

Worked Example 6.1.1 (the four basic single-resistor problems)
(6.1) A $100\,\Omega$ resistor across $12\,\mathrm{V}$: $I = V/R = 12/100 = 0.12\,\mathrm{A}$ and $P = V^2/R = 144/100 = 1.44\,\mathrm{W}$.
(6.2) $1.0\,\mathrm{mA}$ through $4.7\,\mathrm{k\Omega}$: $V = IR = 0.001 \times 4700 = 4.7\,\mathrm{V}$, $P = I^2R = (10^{-3})^2(4700) = 4.7\,\mathrm{mW}$.
(6.3) $24\,\mathrm{V}$ measured across a resistor carrying $50\,\mathrm{mA}$: $R = V/I = 24/0.05 = 480\,\Omega$, $P = VI = 24 \times 0.05 = 1.2\,\mathrm{W}$.
(6.4) $3\,\mathrm{V}$ across $1\,\mathrm{M\Omega}$: $I = 3/10^6 = 3\,\mu\mathrm{A}$, $P = V^2/R = 9/10^6 = 9\,\mu\mathrm{W}$.

6.2 Power ratings

A resistor converts its power to heat, so every resistor has a maximum **power rating** ($\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{2}$, $1\,\mathrm{W}$, …). Choose a part rated for at least twice the expected dissipation.

Worked Example 6.2.1 (largest safe voltage for a $1\,\mathrm{W}$ part)
From $P = V^2/R$, the maximum voltage is $V_{\max} = \sqrt{PR}$. For $P = 1\,\mathrm{W}$: across $2\,\Omega$, $V_{\max} = \sqrt{2} = 1.4\,\mathrm{V}$; across $100\,\Omega$, $10\,\mathrm{V}$; across $3\,\mathrm{k\Omega}$, $54.8\,\mathrm{V}$; across $68\,\mathrm{k\Omega}$, $261\,\mathrm{V}$; across $1\,\mathrm{M\Omega}$, $1000\,\mathrm{V}$. Small resistors reach their power limit at low voltage; large ones can stand hundreds of volts.

Part III

Circuit Building Blocks and Reduction

Chapter 7

Circuit Topology

A **circuit** is any arrangement of elements that lets current flow from a source, through a **load**, and back. Three topologies recur: *series* (one path; the same current through every element), *parallel* (elements across the same two nodes; the same voltage across each), and *combination* (series and parallel nested together).

7.1 Elements by their current–voltage relationship

Lecture 5 introduces each basic element not by a picture but by its I–V **relationship** — what the element fixes and what it lets the rest of the circuit decide. This single framing unifies the whole element set:

Element	Fixes	Free (set by rest of circuit)
Wire	$V_{\mathrm{elem}} = 0$	current (any value)
Open circuit	$I_{\mathrm{elem}} = 0$	voltage (any value)
Resistor	$V = IR$ (Ohm's law)	the operating point on the line
Voltage source	$V_{\mathrm{elem}} = V_S$	current (any value)
Current source	$I_{\mathrm{elem}} = I_S$	voltage (any value)

Key idea The elements come in **dual pairs**. The *wire* (fixes $V = 0$, current free) and the *open circuit* (fixes $I = 0$, voltage free) are duals; so are the *voltage source* (fixes V, current free) and the *current source* (fixes I, voltage free). On an I–V plot the wire is the vertical axis, the open circuit the horizontal axis, the resistor a line of slope $1/R$, and the ideal sources a horizontal or vertical line offset from the origin. Recognising which quantity an element pins down — and which it leaves to the network — is the whole game of setting up circuit equations.

Chapter 8

Resistors in Parallel

Parallel resistors share one voltage and split the current. Because the same V drives each, the branch currents add ($I = \sum V/R_i$), which gives

$$\frac{1}{R_{\text{eq}}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots, \qquad R_1 \parallel R_2 = \frac{R_1 R_2}{R_1 + R_2}. \tag{8.1}$$

The equivalent is always *smaller* than the smallest branch: adding a path can only make it easier for current to flow.

Key idea That the branch currents sum to the total is **Kirchhoff's Current Law** (KCL): charge is conserved at the junction. Applying it to two parallel resistors gives the **current-divider** rule

$$I_1 = I\,\frac{R_2}{R_1 + R_2},$$

handy when you know the total current but not the voltage. The *larger* resistor takes the *smaller* share of current.

Worked Example 8.0.1 (three resistors in parallel, fully worked)

$R_1 = 1\,\mathrm{k\Omega}$, $R_2 = 2\,\mathrm{k\Omega}$, $R_3 = 4\,\mathrm{k\Omega}$ across $24\,\mathrm{V}$. Equivalent: $1/R_{\text{eq}} = 1/1000 + 1/2000 + 1/4000 = 7/4000$, so $R_{\text{eq}} = 571\,\Omega$. Each resistor sees the full $24\,\mathrm{V}$: $I_1 = 24/1000 = 24\,\mathrm{mA}$, $I_2 = 12\,\mathrm{mA}$, $I_3 = 6\,\mathrm{mA}$; total $I = 42\,\mathrm{mA}$ (check: $24/571 = 42\,\mathrm{mA}$). Powers: $P_1 = 24 \cdot 0.024 = 576\,\mathrm{mW}$, $P_2 = 288\,\mathrm{mW}$, $P_3 = 144\,\mathrm{mW}$; total $1.008\,\mathrm{W}$.

Chapter 9

Resistors in Series

Series resistors carry one current and split the voltage. Because the same I flows, the drops add and

$$R_{\mathrm{eq}} = R_1 + R_2 + \cdots \tag{9.1}$$

Key idea That the drops sum to the source voltage is **Kirchhoff's Voltage Law** (KVL): energy is conserved around the loop. Applying it to two series resistors gives the **voltage-divider** rule

$$V_{\mathrm{out}} = V_{\mathrm{in}} \frac{R_2}{R_1 + R_2},$$

worth memorizing — it is the most-used circuit in the course.

Worked Example 9.0.1 (series string, fully worked)

$R_1 = 3.3\,\mathrm{k\Omega}$, $R_2 = 4.7\,\mathrm{k\Omega}$, $R_3 = 10\,\mathrm{k\Omega}$ across $24\,\mathrm{V}$. $R_{\mathrm{eq}} = 18\,\mathrm{k\Omega}$, so $I = 24/18000 = 1.33\,\mathrm{mA}$ through all three. Drops: $V_1 = IR_1 = 4.4\,\mathrm{V}$, $V_2 = 6.27\,\mathrm{V}$, $V_3 = 13.3\,\mathrm{V}$ (sum $\approx 24\,\mathrm{V}$). Powers: $P_1 = I^2 R_1 = 5.9\,\mathrm{mW}$, $P_2 = 8.4\,\mathrm{mW}$, $P_3 = 17.8\,\mathrm{mW}$; total $32\,\mathrm{mW}$.

Worked Example 9.0.2 (designing a divider for a high-impedance load)

An IC needs $5\,\mathrm{V}$ from a $9\,\mathrm{V}$ supply and draws negligible current ($R_{\mathrm{in}} \approx 10\,\mathrm{M\Omega}$). Pick $R_2 = 10\,\mathrm{k\Omega}$; from the divider rule $5 = 9\,R_2/(R_1 + R_2)$, so $R_1 = R_2(9-5)/5 = 10\,\mathrm{k\Omega} \cdot 0.8 = 8\,\mathrm{k\Omega}$. Because the load draws almost nothing, the unloaded divider equation is accurate here.

Chapter 10

Reducing a Complex Resistor Network

Many networks that are neither purely series nor purely parallel can still be collapsed by **circuit reduction**: repeatedly replace the innermost series or parallel cluster by its equivalent until a single resistor remains, find the total current from the source, then work *back outward*, using the divider rules to recover each branch's current and voltage. This handles any *ladder*-style network. Only when a network has no reducible cluster (a bridge, say) must you fall back on the general methods of Part IV.

Chapter 11

Voltage Dividers in Practice: Loading and the 10-Percent Rule

11.1 The loaded divider

The clean divider equation assumes the output drives nothing. A real load R_L hangs in parallel with R_2, lowering the bottom-leg resistance to $R_2 \parallel R_L$ and pulling V_{out} *below* the design value. The divider "sags."

Worked Example 11.1.1 (how much a load pulls a divider down)

$V_{\text{in}} = 9\,\text{V}$, $R_1 = 4.7\,\text{k}\Omega$, $R_2 = 10\,\text{k}\Omega$ gives an unloaded $V_{\text{out}} = 9 \cdot 10/14.7 = 6.12\,\text{V}$. Add a $10\,\text{k}\Omega$ load: the bottom leg becomes $10 \parallel 10 = 5\,\text{k}\Omega$, so $V_{\text{out}} = 9 \cdot 5/(4.7+5) = 4.64\,\text{V}$ — a $1.5\,\text{V}$ sag. A stiff output needs either much smaller divider resistors (more wasted current) or an active buffer.

11.2 The 10-percent rule and the bleeder resistor

When the load *does* draw appreciable current, size the divider with the **10-percent rule**: choose the lower resistor so that the current through it — the **bleeder current** — is about 10% of the load current. That resistor is the **bleeder resistor**. A healthy bleeder current keeps V_{out} stable against load changes while wasting only a little power; too small a bleeder current and the output sags, too large and the divider burns power needlessly.

Worked Example 11.2.1 (sizing a loaded divider)

A $10\,\text{V}$ supply must feed a $3\,\text{V}$, $9.1\,\text{mA}$ load. Bleeder current = 10% of $9.1\,\text{mA} = 0.91\,\text{mA}$. The bleeder resistor carries $3\,\text{V}$: $R_2 = 3/0.00091 \approx 3.3\,\text{k}\Omega$ (use a standard $3.3\,\text{k}\Omega$). The upper resistor carries the load-plus-bleeder current ($9.1 + 0.91 = 10.0\,\text{mA}$) and drops $10 - 3 = 7\,\text{V}$: $R_1 = 7/0.010 = 700\,\Omega$ (use $680\,\Omega$).

11.3 Multiple dividers and split supplies

Several taps can be stacked to power multiple loads from one supply ("multiple voltage divider"); size the bottom bleeder from 10% of the *total* load current, then work upward, adding each load's current to the string as you go. Relocating the ground tap between two resistors yields *both* positive and negative rails from a single source — a split supply with the ground acting as a common return, exactly what audio signals (which swing both ways about 0 V) require.

Key idea Dividers are simple but *unregulated*: any change in load or supply shifts every output. Where a steady voltage under varying load is needed, use an active *voltage-regulator IC*, not a divider.

Chapter 12

Voltage and Current Sources: Ideal versus Real

An **ideal voltage source** holds its terminal voltage fixed for any load current; an **ideal current source** holds its current fixed for any load voltage. Real sources fall short:

- A **real voltage source** is an ideal source in *series* with a small internal resistance r_s; loading it drops voltage across r_s, so the terminal voltage sags: $V_{\text{out}} = V_S\, R_L/(R_L + r_s)$.
- A **real current source** is an ideal source in *parallel* with a large internal resistance.

Key idea **Source transformation:** a real voltage source (V_S in series with r_s) and a real current source ($I_S = V_S/r_s$ in parallel with r_s) are indistinguishable from the outside. This duality is the key that later turns a Thévenin equivalent into a Norton equivalent (Chapter 19).

Chapter 13

Combining Batteries

Batteries in **series** add their voltages while carrying the same current — two $1.5\,\mathrm{V}$ cells make $3.0\,\mathrm{V}$ — which is how you reach a needed supply voltage; chemically, you have doubled the number of charge-pumping reactions in line. Batteries in **parallel** keep the same voltage but add *capacity* and can source more current, extending run time. (Real cells in parallel must be matched, or their small internal resistances let one cell dump current into another.)

Worked Example 13.0.1 (series cells into a load)

Four $1.5\,\mathrm{V}$ AA cells in series make $V_{\mathrm{in}} = 6.0\,\mathrm{V}$. Across $100\,\Omega$: $I = 6/100 = 60\,\mathrm{mA}$, $P = V^2/R = 36/100 = 0.36\,\mathrm{W}$.

Chapter 14

Measuring Voltage, Current, and Resistance

An **ideal voltmeter** has *infinite* input resistance (it draws no current) and connects *in parallel* with the element under test. An **ideal ammeter** has *zero* resistance (it drops no voltage) and connects *in series*. An **ohmmeter** pushes a known current and reads the resulting voltage. In analog meters all three are built around a **galvanometer**, a coil that deflects with current.

Key idea Real meters are not ideal, so they *load* the circuit: a real voltmeter's finite resistance (hundreds of MΩ, not infinite) steals a little current; a real ammeter's small resistance (a fraction of an ohm, not zero) drops a little voltage. The error is worst when the meter's resistance is comparable to the circuit resistance it is measuring — so a voltmeter wants a resistance far *above*, and an ammeter far *below*, the local circuit resistance.

The lecture quantifies the effect with three demonstrations, each showing a few-percent error from meter loading:

Measurement	Actual	Measured	Error
Voltage (across a 1 kΩ leg of a 6 V circuit)	4.0 V	3.9 V	$\sim$3%
Current (in a 10 Ω branch)	0.150 A	0.147 A	$\sim$2%
Resistance (of a 22 Ω part)	22 Ω	23 Ω	$\sim$3%

The lesson: a good meter is not perfect, and for precision work you must know whether the meter's resistance is far enough from the circuit's to keep the loading error below your tolerance.

Chapter 15

Open Circuits, Short Circuits, and the Fuse

A **short circuit** is an unintended near-$0\,\Omega$ path; by $I = V/R$ with $R \to 0$ the current becomes very large, which can overheat wires or destroy parts. An **open circuit** is a break ($R \to \infty$), so no current flows. A **fuse** (Lecture 5, slide 25) is a deliberate weak link rated for a chosen current: it carries normal current but melts — turning itself into an *open* — when the current exceeds its rating, protecting everything downstream. Circuit breakers and transient-voltage suppressors serve the same protective role.

Worked Example 15.0.1 (the three fuse cases, with numbers)

Take a $10\,\mathrm{V}$ source, a $1\,\Omega$ line resistance, a $10\,\Omega$ load, and a $2\,\mathrm{A}$ fuse. *Normal:* $R = 1 + 10 = 11\,\Omega$, so $I = 10/11 = 0.91\,\mathrm{A}$ — below the fuse rating, all is well. *Open circuit* (load disconnects, $R \to \infty$): $I \to 0$. *Short circuit* (load shorts, $R \to 1\,\Omega$): $I = 10/1 = 10\,\mathrm{A}$, far above $2\,\mathrm{A}$, so the fuse blows and becomes an open — exactly the protection it was placed for.

Part IV

Systematic Circuit Analysis

Chapter 16

Kirchhoff's Laws

When reduction stalls, two conservation laws solve *any* network.

Key idea **KCL (junction rule):** at any node, $\sum I_{\text{in}} = \sum I_{\text{out}}$ — conservation of charge.
KVL (loop rule): around any closed loop, $\sum \Delta V = 0$ — conservation of energy.

16.1 The method

Assign a current to each branch and a direction to each (a wrong guess just comes out negative). Write KCL at the independent nodes and KVL around the independent loops, replacing each resistor's voltage with $V_n = I_n R_n$ (Ohm's law). This yields a linear system — as many equations as unknown currents — which you solve by substitution, elimination, or matrices.

Worked Example 16.1.1 (single-loop KVL)

A $10\,\text{V}$ source drives $R_1 = 2\,\text{k}\Omega$ and $R_2 = 3\,\text{k}\Omega$ in series. KVL: $10 - IR_1 - IR_2 = 0 \Rightarrow I = 10/5000 = 2\,\text{mA}$; the drops $IR_1 = 4\,\text{V}$ and $IR_2 = 6\,\text{V}$ sum to the source voltage, as KVL requires.

16.2 Solving the system with determinants (Cramer's rule)

For a multi-loop circuit the algebra grows tedious, so a practical trick is **Cramer's rule**: arrange the equation coefficients into a determinant Δ, form a modified determinant Δ_k for each unknown (replacing the k-th column with the right-hand side), and read off $I_k = \Delta_k/\Delta$. With a calculator or math program that evaluates determinants, you never do the algebra by hand — you fill a grid with coefficients and press "evaluate." This is the fastest hand route for a circuit with three or more unknown currents.

Chapter 17

Nodal Analysis

Lectures 6 and 7 develop **nodal analysis**, the workhorse method, which solves for node *voltages* rather than branch currents (usually fewer unknowns). The course presents it as a fixed seven-step algorithm that never requires cleverness — follow the steps and the equations write themselves:

Step 1. **Pick the reference node** and label it $u = 0\,\mathrm{V}$. Every other node voltage is measured relative to it; conventionally it is the negative terminal of a voltage source, marked with the ground symbol.

Step 2. **Label the remaining nodes** $u_1, u_2, \ldots$ — the unknown voltage at each node relative to the reference.

Step 3. **Label a current** through every element, with an assumed positive direction (a wrong guess simply comes out negative).

Step 4. **Mark voltage-drop directions** with $+/-$ signs on each element using the *passive sign convention.*

Step 5. **Reduce the unknowns**: use each source's known voltage ($u_1 = V_S$, say) and the fact that a node joining only two elements forces equal currents ($i_1 = i_2$) to shrink the variable count.

Step 6. **Write the equations**: KCL at each node, plus each element's I–V relationship ($V_R = iR$), with every element voltage rewritten as a difference of node voltages ($V_{R_1} = u_1 - u_2$).

Step 7. **Solve** the resulting linear system for the node voltages, then back-substitute for any current you need.

Key idea **Passive sign convention:** for a passive element (a resistor), positive current is taken to *enter the + terminal and leave the − terminal.* Following it consistently is what makes the signs in the KCL/Ohm equations come out right without case-by-case reasoning. Applying the full algorithm to a simple two-resistor string reproduces the voltage-divider result $u_2 = V_S\, R_2/(R_1 + R_2)$ — nodal analysis *contains* the divider rule as its simplest case.

Worked Example 17.0.1 (one-unknown node)

A $6\,\mathrm{V}$ source connects through $R_1 = 1\,\mathrm{k\Omega}$ to node A; from A, $R_2 = 2\,\mathrm{k\Omega}$ and $R_3 = 2\,\mathrm{k\Omega}$ each go to ground. KCL at A (currents leaving $= 0$):

$$\frac{V_A - 6}{1\mathrm{k}} + \frac{V_A}{2\mathrm{k}} + \frac{V_A}{2\mathrm{k}} = 0 \;\Rightarrow\; 2(V_A - 6) + V_A + V_A = 0 \;\Rightarrow\; V_A = 3\,\mathrm{V}.$$

The source current is $(6 - 3)/1\mathrm{k} = 3\,\mathrm{mA}$.

Key idea **The touchpad demo board.** The lecture's capstone example is the demo board's resistive *touch panel*: a 3×3 mesh of nine nodes (S_1–S_9) joined by twelve resistors, driven at 3.3 V with one corner tied to ground. No amount of series/parallel reduction collapses this mesh, so it is solved by exactly the seven-step algorithm above — KCL at every node, each branch current written by Ohm's law, and the resulting linear system solved for the nine node voltages. This is why nodal analysis matters: a real interface (where you touch the panel changes a resistance and shifts the node voltages the ADC reads) is a mesh, not a ladder, and only the systematic method reaches it.

Chapter 18

Superposition

In a *linear* circuit with several independent sources, the current (or voltage) in any branch equals the **sum** of the contributions of each source acting alone. To silence the others: replace each inactive *voltage* source with a short (0 V), and each inactive *current* source with an open (0 A). Solve the simple one-source circuits and add the partial results.

> **Key idea** Superposition works only because linear elements obey $V = IR$ with constant R — doubling all sources doubles every response. It fails the instant a nonlinear element (a diode) appears. It is rarely the quickest method on its own, but it is the idea underneath Thévenin's and Norton's theorems.

Chapter 19

Thévenin's and Norton's Theorems

When only one pair of terminals matters — the current through one load, say — collapsing the entire rest of the network into a two-element equivalent is far faster than a full Kirchhoff solve.

Key idea Any linear two-terminal network behaves, at its terminals, like either equivalent:

- **Thévenin:** one voltage source V_{TH} in *series* with a resistance R_{TH}.
- **Norton:** one current source I_{N} in *parallel* with the same R_{TH}.

Treat the network as a *black box* at terminals A–B and find:

$$\begin{aligned} V_{\text{TH}} &= \text{open-circuit voltage across A–B (load removed)}, \\ I_{\text{N}} &= \text{short-circuit current across A–B}, \\ R_{\text{TH}} &= \text{resistance seen at A–B with every independent source zeroed} \\ &\quad\ \text{(voltage sources shorted, current sources opened)}, \end{aligned}$$

related by $V_{\text{TH}} = I_{\text{N}}\, R_{\text{TH}}$. The load current then follows from $I = V_{\text{TH}}/(R_{\text{TH}} + R_L)$.

19.1 Finding the equivalent

Remove the load and open the two terminals of interest. V_{TH} is the voltage now appearing across the gap (often just a divider calculation). R_{TH} is what an ohmmeter would read into the terminals once all internal DC sources are replaced by shorts. I_{N} is the current that flows if you instead *short* the terminals. Any Thévenin equivalent converts to Norton (and back): the resistance is the same, in series with the voltage source or in parallel with the current source. The Thévenin voltage equals the no-load voltage of the Norton form, and the Norton current equals the short-circuit current of the Thévenin form. When the terminals of interest sit *inside* a network (across some R_{LOAD}), remove that resistor first to expose the terminals, find the equivalent, then reattach the resistor and apply Ohm's law.

Worked Example 19.1.1 (Thévenin equivalent of a loaded divider)

Divider: $V_{\text{in}} = 9\,\text{V}$, $R_1 = 4.7\,\text{k}\Omega$ (top), $R_2 = 10\,\text{k}\Omega$ (bottom), output across R_2. *Open-circuit voltage:* $V_{\text{TH}} = 9 \cdot 10/(4.7 + 10) = 6.12\,\text{V}$. *Thévenin resistance* (short the 9 V source): R_1 and R_2 are then in parallel, $R_{\text{TH}} = 4.7 \,\|\, 10 = 3.20\,\text{k}\Omega$. So the whole divider behaves like a 6.12 V source with 3.20 kΩ in series. A 10 kΩ load now sees $V_{\text{out}} = 6.12 \cdot 10/(10 + 3.20) = 4.64\,\text{V}$ —

instantly reproducing the loaded-divider result from Chapter 11.

Worked Example 19.1.2 (Norton form, and batteries in parallel)

Converting the previous result: $I_{\mathrm{N}} = V_{\mathrm{TH}}/R_{\mathrm{TH}} = 6.12/3200 = 1.91\,\mathrm{mA}$ in parallel with $3.20\,\mathrm{k}\Omega$; check $I_{\mathrm{N}}R_{\mathrm{TH}} = 6.12\,\mathrm{V} = V_{\mathrm{TH}}$. The same machinery models real cells in parallel: two $1.5\,\mathrm{V}$ cells, each with internal resistance $0.2\,\Omega$, present a Thévenin source of $1.5\,\mathrm{V}$ with $R_{\mathrm{TH}} = 0.2 \,\|\, 0.2 = 0.1\,\Omega$ — the halved internal resistance is exactly why paralleling cells raises current capacity.

Part V

Energy-Storage Elements

Chapter 20

Capacitors

A **capacitor** stores energy in the electric field between two plates. Its defining relations are

$$Q = CV, \qquad I = C\,\frac{dV}{dt}, \qquad E = \tfrac{1}{2}CV^2, \tag{20.1}$$

with C in farads. The capacitance $C = Q/V$ is the charge stored per volt applied. Because $I = C\,dV/dt$, current flows only while the voltage is *changing*; a capacitor therefore **opposes sudden changes in voltage** — its voltage cannot jump. Charging through a series resistor, it follows an exponential with time constant $\tau = RC$. In the water analogy a capacitor is a tank sealed by a stretchy rubber membrane: water (charge) pushes the membrane, storing energy, but no water passes through once it is taut. Common types include *fixed*, *polarized* (electrolytic), *variable*, and *trimmer* capacitors.

Worked Example 20.0.1 (RC time constant)
$R = 10\,\mathrm{k\Omega}$, $C = 1\,\mu\mathrm{F}$: $\tau = RC = 10^4 \times 10^{-6} = 10\,\mathrm{ms}$. After a voltage step, the capacitor reaches $\sim$63% of the final value in one τ and is $> 99\%$ charged after $5\tau = 50\,\mathrm{ms}$.

20.1 Capacitors in series and parallel

Crucially, capacitors combine **opposite** to resistors:

$$\text{parallel: } C_{\text{eq}} = C_1 + C_2 + \cdots, \qquad \text{series: } \frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots \tag{20.2}$$

Key idea Placing capacitors in *parallel* **increases** the total capacitance (the plate areas add) but limits the assembly's maximum voltage rating to that of the *smallest*-rated part. Placing them in *series* **decreases** the capacitance but *raises* the voltage rating (the applied voltage divides among them). This is the exact mirror image of resistors — worth memorising as "capacitors add in parallel."

Chapter 21

Inductors

An **inductor** stores energy in the magnetic field around a coil:

$$V = L\,\frac{dI}{dt}, \qquad I = \frac{1}{L}\int V\,dt, \qquad E = \tfrac{1}{2}LI^2, \tag{21.1}$$

with L in henries. By duality with the capacitor it **opposes sudden changes in current** — its current cannot jump — and in series with a resistor settles with time constant $\tau = L/R$. In the water analogy an inductor is a heavy paddle-wheel or flywheel: it resists starting and stopping the flow, storing energy in its motion. Types include *air-core*, *iron-core* (magnetic), *adjustable*, and *ferrite-bead* inductors.

21.1 Inductors in series and parallel

Inductors combine **like resistors** (and therefore opposite to capacitors):

$$\text{series: } L_{\text{eq}} = L_1 + L_2 + \cdots, \qquad \text{parallel: } \frac{1}{L_{\text{eq}}} = \frac{1}{L_1} + \frac{1}{L_2} + \cdots \tag{21.2}$$

Key idea Capacitor and inductor are duals: swap $V \leftrightarrow I$ and $C \leftrightarrow L$ and each element's equations become the other's. The capacitor resists voltage change; the inductor resists current change. These two elements, added to the resistor, complete the set needed for the time-dependent circuits of the next unit.

Appendix A

Consolidated Formula Sheet

Quantity / law	Equation
Current	$I = dQ/dt, \quad 1\,\mathrm{A} = 1\,\mathrm{C/s}$
Voltage	$V = U/q, \quad 1\,\mathrm{V} = 1\,\mathrm{J/C} = 1\,\mathrm{W/A}$
Power (generalized)	$P = VI$
Resistor power law	$P = VI = I^2R = V^2/R$
Resistance (definition)	$R = V/I$
Ohm's law (ohmic only)	$V = IR$
Current density	$J = I/A$
Resistivity / conductivity	$R = \rho L/A, \quad \sigma = 1/\rho$
Temperature coefficient	$\rho = \rho_0[1 + \alpha(T - T_0)]$
Series resistors	$R_{\mathrm{eq}} = R_1 + R_2 + \cdots$
Parallel resistors	$1/R_{\mathrm{eq}} = \sum 1/R_i, \quad R_1 \parallel R_2 = \dfrac{R_1R_2}{R_1 + R_2}$
Voltage divider	$V_{\mathrm{out}} = V_{\mathrm{in}}\, R_2/(R_1 + R_2)$
Current divider	$I_1 = I\, R_2/(R_1 + R_2)$
Real voltage source	$V_{\mathrm{out}} = V_S\, R_L/(R_L + r_s)$
10-percent rule	$I_{\mathrm{bleeder}} \approx 0.1\, I_{\mathrm{load}}$
KCL / KVL	$\sum I_{\mathrm{in}} = \sum I_{\mathrm{out}} \quad / \quad \sum \Delta V = 0$
Cramer's rule	$I_k = \Delta_k/\Delta$
Thévenin / Norton	$V_{\mathrm{TH}} = I_{\mathrm{N}}R_{\mathrm{TH}}, \quad I_{\mathrm{load}} = V_{\mathrm{TH}}/(R_{\mathrm{TH}} + R_L)$
Capacitor	$Q = CV, \quad I = C\,dV/dt, \quad E = \frac{1}{2}CV^2, \quad \tau = RC$
Capacitors (parallel/series)	$C_{\mathrm{eq}} = \sum C_i \quad / \quad 1/C_{\mathrm{eq}} = \sum 1/C_i$
Inductor	$V = L\,dI/dt, \quad E = \frac{1}{2}LI^2, \quad \tau = L/R$
Inductors (series/parallel)	$L_{\mathrm{eq}} = \sum L_i \quad / \quad 1/L_{\mathrm{eq}} = \sum 1/L_i$